

GhostIntel — Master Blueprint

Official Website: ghintel.web.id

Project Type: OSINT Framework

Primary Language: Python

Interface: CLI + Web UI

Target Environment: Linux, Windows, Termux

Architecture: Modular + Service/Core Architecture

Database: SQLite sebagai default local backend

Reporting: JSON / Markdown / HTML / TXT

Notification: Telegram Bot

Localization: Indonesian + English

Update Source: Official GitHub

Official Documentation: ghintel.web.id

1. Visi GhostIntel

GhostIntel diarahkan menjadi:

A modular OSINT framework for collecting, validating, correlating, and reporting publicly available information.

Fokus utamanya bukan:

“sebanyak apa hasil yang bisa ditampilkan?”

Tetapi:

“sebanyak apa informasi yang benar-benar bisa dibuktikan?”

Ini penting terutama untuk username.

Kalau sebuah platform tidak bisa memastikan username benar-benar ada, GhostIntel **tidak boleh mengarang** **FOUND** hanya karena request menghasilkan HTTP 200.

2. Prinsip Utama GhostIntel

2.1 Accuracy First

Setiap result harus memiliki status yang jelas.

```
FOUND
NOT_FOUND
NOT_ACTIVE
UNKNOWN
ERROR
TIMEOUT
RATE_LIMITED
BLOCKED
```

Contoh:

```
Instagram
Username : example
Status   : FOUND
```

atau:

```
TikTok
Username : example
Status   : NOT_FOUND
```

atau:

```
Twitter/X
Username : example
Status   : UNKNOWN
Reason    : Request blocked / verification unavailable
```

UNKNOWN tidak boleh diubah menjadi NOT_FOUND.

3. Target Environment

GhostIntel harus dianggap sebagai **CLI-first application**.

Bukan aplikasi desktop yang kebetulan bisa dibuka melalui terminal.

Target:

```
Windows
Linux
Termux
```

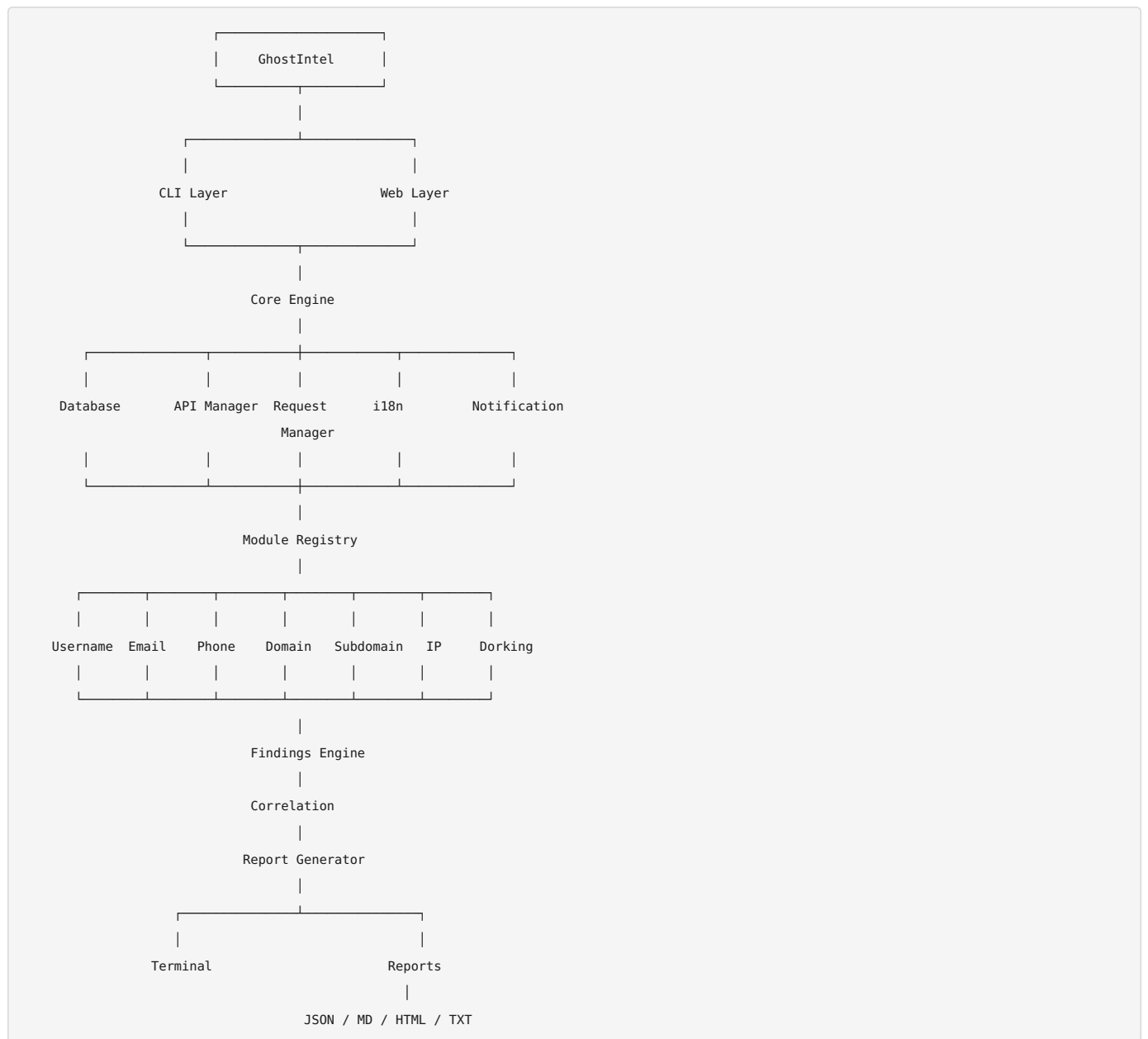
Termux harus menjadi environment resmi yang diperhatikan sejak awal development.

Hindari

- dependency Windows-only
- GUI-only libraries
- path hardcoded
- executable-specific assumptions
- native dependency yang tidak perlu
- fitur yang hanya bekerja di desktop

4. Arsitektur Besar

Struktur high-level:



5. Struktur Folder Final

Blueprint struktur repository:

```
GhostIntel/
|
├─ ghostintel.py
├─ pyproject.toml
├─ requirements.txt
├─ requirements-dev.txt
├─ README.md
├─ CHANGELOG.md
├─ LICENSE
├─ CONTRIBUTING.md
├─ SECURITY.md
├─ .gitignore
|
├─ core/
|   └─ __init__.py
```

```
|   ├── engine.py
|   ├── config.py
|   ├── database.py
|   ├── request_manager.py
|   ├── api_manager.py
|   ├── rate_limiter.py
|   ├── cache.py
|   ├── proxy.py
|   ├── logger.py
|   ├── debug.py
|   ├── bug_reporter.py
|   ├── i18n.py
|   ├── module_registry.py
|   ├── findings.py
|   ├── correlation.py
|   ├── monitor.py
|   ├── notifier.py
|   ├── updater.py
|   └── exceptions.py
|
└── modules/
    ├── __init__.py
    ├── username.py
    ├── email.py
    ├── phone.py
    ├── domain.py
    ├── subdomain.py
    ├── dns.py
    ├── ip.py
    ├── asn.py
    ├── certificate.py
    ├── breach.py
    ├── reputation.py
    └── dorking.py
|
└── sources/
    ├── __init__.py
    ├── username/
    ├── email/
    ├── phone/
    ├── domain/
    └── common/
|
└── dorks/
    ├── level1.yaml
    └── level2.yaml
|
└── locales/
    ├── id.json
    └── en.json
|
└── reports/
    ├── generator.py
    ├── schema.py
    └── templates/
        ├── html/
        ├── markdown/
        └── txt/
|
└── notifications/
    ├── __init__.py
    └── telegram.py
```

```
|
├─ web/
|   ├─ app.py
|   ├─ routes/
|   ├─ templates/
|   └─ static/
|
├─ tests/
|   ├─ unit/
|   ├─ integration/
|   ├─ e2e/
|   └─ fixtures/
|
├─ examples/
|   └─ config.example.yaml
|
└─ docs/
    ├─ installation.md
    ├─ termux.md
    ├─ configuration.md
    ├─ modules.md
    ├─ api.md
    ├─ reports.md
    ├─ dorking.md
    ├─ troubleshooting.md
    └─ development.md
```

6. CLI Architecture

CLI menjadi interface utama.

Contoh:

```
ghostintel
```

Menampilkan:

GhostIntel v3.x

Usage:

ghostintel [command] [options]

Commands:

username	Username intelligence
email	Email intelligence
phone	Phone intelligence
domain	Domain intelligence
subdomain	Subdomain enumeration
ip	IP intelligence
dork	Dorking Engine
scan	Combined OSINT scan
report	Report management
monitor	Monitoring
api	API management
config	Configuration
debug	Debug tools
bug	Bug reporting
update	Update GhostIntel
help	Help
version	Show version

7. Help System

Help harus benar-benar berguna.

Contoh:

```
ghostintel --help
```

atau:

```
ghostintel username --help
```

Output jangan terlalu formal.

Bahasa dibuat:

professional + casual.

Contoh Indonesian:

Cek username di berbagai platform.

Usage:

ghostintel username <username>

Options:

--deep

Ambil informasi profile yang tersedia.

--save

Simpan hasil lengkap ke report.

--format

Pilih format report.

--json

Output JSON.

--telegram

Kirim ringkasan ke Telegram.

8. Official Website Note pada Help

Ini masuk requirement khusus kita.

Di bagian bawah banner/help utama:

Official Website:

<https://www.ghintel.web.id>

Kunjungi website resmi GhostIntel untuk dokumentasi,
update, panduan, dan informasi terbaru.

English:

Official Website:

<https://www.ghintel.web.id>

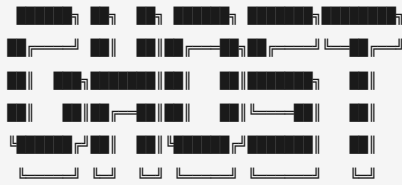
Visit the official GhostIntel website for documentation,
updates, guides, and the latest information.

Karena seluruh text harus masuk i18n, **URL-nya tetap sama**, sedangkan deskripsinya berubah berdasarkan language.

9. Banner

Banner jangan terlalu besar.

Contoh:



GhostIntel
OSINT Intelligence Framework

Version 3.x

Setelah banner langsung:

Official Website:
<https://www.ghintel.web.id>

10. Language System

Bahasa harus **global**.

Bukan hanya CLI.

Yang mengikuti language:

CLI
Web UI
Errors
Warnings
Reports
Telegram
Bug reports
Updater
Help
Module output
Database-generated messages

11. i18n Architecture

```
locales/  
├─ id.json  
└─ en.json
```

Contoh:

```
{  
  "app.name": "GhostIntel",  
  "help.website": "Kunjungi website resmi GhostIntel untuk dokumentasi, update, panduan, dan informasi terbaru.",  
  "scan.start": "Mulai scan...",  
  "scan.complete": "Scan selesai.",  
  "result.found": "Ditemukan",  
  "result.not_found": "Tidak ditemukan",  
  "result.unknown": "Belum bisa dikonfirmasi"  
}
```

English:

```
{
  "app.name": "GhostIntel",
  "help.website": "Visit the official GhostIntel website for documentation, updates, guides, and the latest information.",
  "scan.start": "Starting scan...",
  "scan.complete": "Scan completed.",
  "result.found": "Found",
  "result.not_found": "Not found",
  "result.unknown": "Could not be confirmed"
}
```

12. Username Intelligence Engine

Ini salah satu core terbesar.

Command:

```
ghostintel username ruyynn
```

Basic:

```
ghostintel username ruyynn
```

Deep:

```
ghostintel username ruyynn --deep
```

Save:

```
ghostintel username ruyynn --deep --save
```

13. Username Result Model

Setiap platform:

```
Platform
Username
Status
Profile URL
Display Name
Bio
Avatar
Followers
Following
Posts
Videos
Account Created
Last Seen
Verification
Evidence
Source
Confidence
```

Tapi:

jangan mengarang data.

Jika platform tidak menyediakan followers:

```
Followers: NULL
```

bukan:

```
Followers: 0
```

karena `0` berarti benar-benar nol.

14. Truthful Detection

Engine harus platform-aware.

Jangan:

```
if response.status_code == 200:
    found = True
```

Karena banyak website:

- soft 404
- redirect
- login wall
- anti-bot
- Cloudflare
- generic 200 page

Jadi perlu:

```
Platform Adapter
  ↓
Request
  ↓
Response
  ↓
Fingerprint
  ↓
Validation
  ↓
Evidence
  ↓
Status
```

15. Username Status

FOUND

Account berhasil diverifikasi.

NOT_FOUND

Account tidak ditemukan dan bukti cukup.

NOT_ACTIVE

Account/profile pernah/terindikasi ada tetapi sekarang tidak aktif atau unavailable dengan indikator yang cukup.

UNKNOWN

Tidak dapat diverifikasi.

ERROR

Internal/module error.

TIMEOUT

Request timeout.

RATE_LIMITED

Platform membatasi request.

BLOCKED

Request diblokir.

16. Deep Username Scan

Jika:

```
50 platforms
```

dan user memilih save:

```
Terminal:

[1/50] Instagram      FOUND
[2/50] Facebook      NOT_FOUND
[3/50] TikTok        FOUND
...
[10/50] Discord       UNKNOWN

Showing preview: 10/50
Full results saved.
```

Kalau **tidak save**:

```
Showing preview: 25/50
```

Jadi:

```
SAVE MODE
→ terminal 10
→ file 50

NO SAVE
→ terminal 25
```

17. Deep Profile Preview

Contoh:

Instagram

Username : example
Status : FOUND
Profile : <https://instagram.com/example>

Display Name: Example User

Bio : ...

Followers : 12.4K

Following : 421

Posts : 83

Account Age : Unknown

Avatar : Available

Evidence:

Profile page verified

Data hanya ditampilkan jika benar-benar tersedia.

18. Complete Report

Report:

reports/
└─ username-example-2026-09-28.json

Isinya semua result:

1/50
2/50
3/50
...
50/50

Tidak dipotong.

19. Email Intelligence

Email module diperluas.

Basic:

Syntax
Domain
MX
SPF
DMARC
Disposable
Role account
Domain reputation

Advanced:

```
Provider API
Breach lookup
Reputation
Additional intelligence
```

20. API Integration — Prinsip Penting

Kalau user memasukkan API:

```
ghostintel api set <provider>
```

GhostIntel **harus benar-benar menggunakan API tersebut.**

Tidak boleh:

```
API Key: configured
```

tetapi module tidak pernah memanggil API.

21. API Manager

Semua API melalui:

```
core/api_manager.py
```

Fungsi:

```
store_key()
get_key()
delete_key()
test_key()
list_providers()
provider_status()
provider_capabilities()
```

CLI:

```
ghostintel api list
ghostintel api test
ghostintel api set
ghostintel api remove
```

22. API Capability System

Misalnya:

```
HIBP
├─ breach lookup
└─ email intelligence

Provider X
├─ reputation
└─ enrichment
```

Jika API tidak dikonfigurasi:

```
Advanced email intelligence unavailable.
```

Jika API tersedia:

```
Advanced email intelligence enabled.
```

23. API Security

API key:

jangan pernah masuk log.

Jangan:

```
API_KEY=123456789
```

di debug.

Log:

```
API_KEY=[REDACTED]
```

Prioritas storage:

```
OS Keyring
  ↓
Encrypted Local Storage
  ↓
Environment Variable
  ↓
Config
```

24. Database

Default:

SQLite.

Lokasi:

```
~/.ghostintel/ghostintel.db
```

Database dipakai untuk:

```
configuration
API credentials metadata
scan history
results
findings
monitoring
notifications
bug reports
update metadata
cache metadata
```

25. Database Schema

Minimal:

```
settings
api_keys
targets
scans
scan_results
platform_results
findings
sources
monitor_targets
monitor_snapshots
notifications
bug_reports
update_cache
```

26. Scan History

User bisa melihat:

```
ghostintel scan history
```

Contoh:

ID	Target	Type	Status	Date
001	example	username	COMPLETE	2026-09-28
002	example.com	domain	COMPLETE	2026-09-27
003	user@email	email	ERROR	2026-09-26

27. Request Manager

Semua network request harus lewat:

```
core/request_manager.py
```

Jangan setiap module membuat sistem request sendiri.

Request manager menangani:

```
HTTP
Timeout
Retry
Headers
Proxy
Cache
Rate Limit
429
5xx
Connection errors
Logging
```

Architecture:

```
Module
  ↓
Request Manager
  ↓
Rate Limiter
  ↓
Cache
  ↓
Proxy
  ↓
HTTP
```

28. Global Rate Limiter

Rate limiter bukan per-module secara acak.

Gunakan:

```
Global Rate Limiter
+
Provider Rate Limiter
```

Contoh:

```
Instagram → limit A
GitHub    → limit B
Telegram  → limit C
Provider API → limit D
```

29. Cache

Cache berguna agar GhostIntel tidak mengulang request yang sama terus-menerus.

Contoh:

```
username + platform + timestamp
```

Cache TTL configurable.

30. Proxy Support

Optional:

```
ghostintel --proxy http://127.0.0.1:8080
```

atau config:

```
proxy:
  enabled: false
  url: null
```

Semua module harus menggunakan Request Manager sehingga proxy benar-benar berlaku secara global.

31. Domain Intelligence

Domain module diperluas.

Command:

```
ghostintel domain example.com
```

Information:

```
WHOIS
DNS
DNSSEC
A
AAAA
MX
NS
TXT
SPF
DMARC
DKIM
HTTP
HTTPS
Redirect
Headers
SSL
Certificate
Technology
Hosting
IP
ASN
Domain age
Subdomains
```

32. Subdomain Intelligence

Basic GhostIntel:

```
ghostintel subdomain example.com
```

Output:

```
www.example.com
api.example.com
mail.example.com
dev.example.com
```

Tambahkan:

```
source
resolved IP
status
HTTP
HTTPS
technology
```

Untuk functionality yang lebih advanced, architecture dibuat supaya bisa diintegrasikan dengan project **sub-intel** tanpa membuat GhostIntel bergantung penuh kepadanya.

Catatan: detail implementasi repository tersebut jangan diasumsikan sampai source-nya benar-benar tersedia/terverifikasi.

33. Dorking Engine

Dark Web dihapus.

Tidak sekadar disembunyikan.

Hapus dari:

```
CLI
Registry
Config
Docs
Web
Help
Module list
```

Diganti:

Dorking Engine

34. Dorking Level 1

Level 1 untuk pencarian dasar.

Contoh template:

```
site:{domain}
site:{domain} filetype:pdf
site:{domain} filetype:doc
site:{domain} inurl:login
site:{domain} inurl:admin
site:{domain} inurl:api
```

User:

```
ghostintel dork example.com
```

Pilih:

```
1. Level 1
2. Level 2
3. Custom
```

35. Dorking Level 2

Lebih kompleks.

Contoh:

```
site:{domain} (filetype:pdf OR filetype:doc OR filetype:docx)
```

```
site:{domain} inurl:(api OR dev OR staging OR test)
```

```
site:{domain} intitle:(index.of OR login OR dashboard)
```

Template disimpan:

```
dorks/  
├─ level1.yaml  
└─ level2.yaml
```

Jadi penambahan dork tidak harus mengubah Python code.

36. Dorking Output

GhostIntel tidak perlu menjadi search engine.

Fungsinya:

```
generate  
organize  
categorize  
export
```

Contoh:

```
Dorking Engine  
  
Target:  
example.com  
  
Level:  
2  
  
Generated queries:  
01. site:example.com ...  
02. site:example.com ...  
03. site:example.com ...
```

Bisa generate search URL untuk search engine yang dipilih.

37. Scope Dorking

Tetap fokus pada:

public information discovery.

Bukan:

```
exploit automation  
credential theft  
unauthorized access  
attack automation
```

38. Phone Intelligence

Tetap dipertahankan.

Informasi:

Country
Country Code
Region
Carrier
Line Type
Timezone
Validation
Formatting
Possible reputation

Jika API tersedia:

API enrichment

39. IP Intelligence

Module:

ghostintel ip 1.1.1.1

Informasi:

IP
Version
Country
Region
City
ASN
ISP
Organization
Reverse DNS
Hostname
Reputation

40. ASN Module

ghostintel asn AS13335

Informasi:

ASN
Organization
Country
Prefixes
Related IP ranges

41. DNS Module

Dedicated module:

```
ghostintel dns example.com
```

Record:

```
A
AAAA
MX
NS
TXT
CNAME
CAA
SOA
SRV
```

42. Certificate Module

SSL/TLS:

```
Issuer
Subject
SAN
Valid From
Valid Until
Serial
Fingerprint
Protocol
```

Bisa membantu menemukan relationship domain/subdomain.

43. Reputation Module

Modular provider architecture:

```
Reputation
├─ Provider A
├─ Provider B
└─ Provider C
```

Kalau provider tidak aktif:

```
Unavailable
```

Jangan fake result.

44. Breach Module

Breach checking hanya menggunakan provider yang benar-benar terintegrasi.

Output:

Breached: YES/NO/UNKNOWN

Source

Date

Affected service

Available metadata

Tidak menampilkan data sensitif yang tidak seharusnya diekspos.

45. Findings Engine

Semua module menghasilkan standardized finding.

Contoh:

```
{  
  "type": "username",  
  "target": "example",  
  "source": "instagram",  
  "status": "FOUND",  
  "confidence": "HIGH",  
  "evidence": "profile_verified"  
}
```

Dengan begitu semua module bisa menggunakan reporting engine yang sama.

46. Evidence System

Setiap finding sebisa mungkin punya:

Source

URL

Timestamp

Evidence

Validation

Confidence

Contoh:

```
Status      : FOUND  
Confidence  : HIGH  
Source      : Instagram  
Evidence    : Public profile verified  
Checked     : 2026-09-28 11:32:41
```

47. Correlation Engine

Setelah beberapa module selesai, GhostIntel bisa menghubungkan hasil.

Contoh:

```
Username
↓
Instagram
↓
Profile URL
↓
Website
↓
Domain
↓
IP
↓
ASN
```

atau:

```
Email
↓
Domain
↓
MX
↓
Company
```

Correlation tidak boleh mengubah dugaan menjadi fakta.

Harus diberi label:

```
Confirmed
Possible
Unconfirmed
```

48. Combined Scan

Command:

```
ghostintel scan example
```

atau:

```
ghostintel scan --username example --email user@example.com --domain example.com
```

Engine:

```
Target
↓
Target Analyzer
↓
Module Selection
↓
Parallel/Controlled Execution
↓
Findings
↓
Correlation
↓
Report
↓
Notification
```

49. Parallel Execution

Untuk speed:

```
Username ┘
Email ───┘
Domain ───┘→ Engine
IP ───┘
Phone ───┘
```

Tetapi tetap tunduk kepada:

```
rate limit
timeout
provider limit
resource limit
```

Jangan membuat 500 request bersamaan hanya karena bisa.

50. Report Engine

Format:

```
JSON
Markdown
HTML
TXT
```

Command:

```
ghostintel report
```

atau:

```
ghostintel username example --save --format html
```

51. Report Structure

GhostIntel Report

Target
Scan Type
Started
Completed
Duration

Summary

Found
Not Found
Not Active
Unknown
Errors
Rate Limited
Blocked

Detailed Results

Platform 1
Platform 2
Platform 3
...

52. HTML Report

HTML harus lebih readable.

Section:

Header
Summary
Statistics
Findings
Sources
Evidence
Technical Details
Errors
Scan Metadata

53. Telegram Integration

Telegram Bot:

Bot Token
Chat ID

disimpan melalui config/API manager.

Command:

ghostintel telegram test

54. Telegram Notification Events

Contoh:

```
SCAN_STARTED
SCAN_COMPLETED
SCAN_FAILED
HIGH_RISK_FINDING
MONITOR_CHANGE
API_ERROR
UPDATE_AVAILABLE
SYSTEM_ERROR
```

55. Telegram Anti-Spam

Jangan:

```
50 platform
→ 50 Telegram messages
```

Lebih baik:

```
Scan complete.

Target: example

Found: 12
Not Found: 21
Unknown: 9
Blocked: 5
Errors: 3

Report:
username-example.html
```

Jika perlu detail:

```
Batch 1/3
Batch 2/3
Batch 3/3
```

Dengan:

```
rate limit
queue
deduplication
batching
cooldown
```

56. Monitoring

GhostIntel bisa menyimpan target:

```
ghostintel monitor add example.com
```

Kemudian:

```
Monitor
↓
Periodic scan
↓
Snapshot
↓
Compare
↓
Change detected
↓
Telegram
```

Perubahan:

```
New subdomain
DNS changed
Certificate changed
Website changed
Username status changed
```

57. Debug System

Command:

```
ghostintel debug
```

atau:

```
ghostintel --debug
```

Output log:

```
~/ghostintel/logs/
```

58. Debug Logging

Level:

```
ERROR
WARNING
INFO
DEBUG
TRACE
```

Default:

```
INFO
```

Debug mode:

```
DEBUG
```

Jangan spam terminal dengan debug information secara default.

59. Secret Redaction

Debug system wajib melakukan:

```
API key → REDACTED
Telegram token → REDACTED
Password → REDACTED
Authorization header → REDACTED
Cookie → REDACTED
```

60. Crash Handler

Jika crash:

```
GhostIntel encountered an unexpected error.

Bug ID:
GI-20260928-XXXX
```

Lalu membuat:

```
debug bundle
```

berisi:

```
version
OS
Python
module
command
error
traceback
sanitized config
relevant logs
```

61. Bug Reporting

Command:

```
ghostintel bug
```

User dapat:

```
What happened?
What were you doing?
Expected result?
Actual result?
```

Optional:

```
Attach debug report?
```

62. Bug Report Format

Subject:

[GhostIntel Bug] Username Scan - GI-XXXXXXX

Body:

GhostIntel Version:

OS:

Python:

Module:

Command:

Expected:

Actual:

Error:

Bug ID:

Additional Notes:

Data target sensitif **tidak otomatis ikut terkirim**.

63. Web Bug Report

Web UI:

Report a Bug

Name

Email

GhostIntel Version

Module

What happened?

Expected result

Actual result

Bug ID

Attach Debug Report

Screenshot

Submit

64. Configuration

Contoh:

```
language: id

database:
  path: ~/.ghostintel/ghostintel.db

network:
  timeout: 15
  retries: 2

rate_limit:
  enabled: true

proxy:
  enabled: false
  url: null

telegram:
  enabled: false
  chat_id: null

reports:
  default_format: html

debug:
  enabled: false
```

Secret jangan hardcoded.

65. GhostIntel Home Directory

```
~/.ghostintel/
|
├─ ghostintel.db
├─ config.yaml
├─ cache/
├─ logs/
├─ debug/
├─ reports/
└─ exports/
```

Config/database user **tidak boleh dihancurkan ketika update aplikasi**.

66. Updater

GhostIntel mengecek official GitHub.

Contoh:

```
Current Version:
3.0.0

Latest Version:
3.5.0

Update available.
```

67. Update Manifest

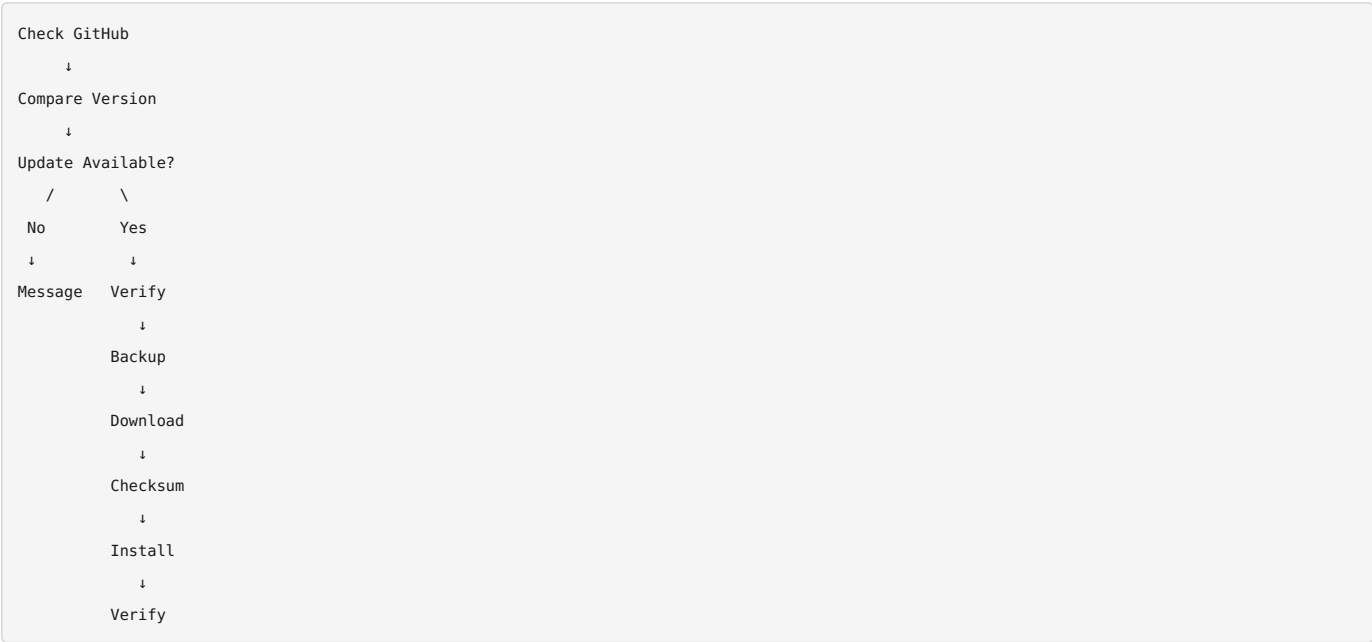
Misalnya:

```
version.json
```

berisi:

```
{
  "version": "3.5.0",
  "minimum_python": "3.10",
  "release_date": "2026-...",
  "download": "...",
  "checksum": "...",
  "notes": "...",
}
```

68. Update Flow



69. Auto Update

Default:

```
OFF
```

Lebih aman.

User dapat:

```
ghostintel update
```

Untuk automated update bisa disediakan optional flag.

70. Config Preservation

Update tidak boleh menghancurkan:

```
~/.ghostintel/config.yaml
~/.ghostintel/ghostintel.db
API configuration
reports
cache
```

71. No Update Message

Ketika tidak ada update:

Indonesian

```
Maaf, saat ini belum ada update terbaru.

Kalau kamu punya ide atau masukan untuk update GhostIntel berikutnya,
silakan hubungi saya melalui Gmail.

Terima kasih sudah menggunakan GhostIntel.

-Ruyynn
```

English

```
Sorry, there is no new update available right now.

If you have ideas or suggestions for the next GhostIntel update,
feel free to contact me via Gmail.

Thank you for using GhostIntel.

-Ruyynn
```

72. Test Suite

Wajib ada.

```
tests/
├─ unit/
├─ integration/
├─ e2e/
└─ fixtures/
```

73. Unit Tests

Test:

```
Username detector
Email validator
Phone parser
Domain parser
DNS parser
Dork generator
API manager
Request manager
Rate limiter
Cache
Database
i18n
Report generator
Updater
Bug reporter
Notifier
```

74. HTTP Fixtures

Test cases:

```
HTTP 200
HTTP 404
HTTP 403
HTTP 429
HTTP 500
Timeout
Connection error
Cloudflare
Invalid JSON
Redirect
Soft 404
```

Ini sangat penting untuk username engine.

75. Integration Tests

Contoh:

```
Module
↓
Request Manager
↓
API Manager
↓
Database
↓
Report
```

Harus dites sebagai sistem.

76. End-to-End Test

Contoh:

```
ghostintel username testuser --save
```

Expected:

```
scan starts
modules run
results generated
database updated
report created
```

77. Language Tests

Test bahwa:

```
ID
```

dan:

```
EN
```

tidak meninggalkan string hardcoded.

Contoh test:

```
switch language
→ CLI changes
→ report changes
→ notification changes
```

78. API Tests

Jika API provider configured:

```
ghostintel api test
```

harus benar-benar menguji koneksi.

Expected:

```
Provider: X
Status: CONNECTED
Capabilities:
- breach
- reputation
```

Jika invalid:

```
Status: INVALID
Reason: Authentication failed
```

79. Termux Test

Minimum:

```
Python installation
CLI
SQLite
HTTP requests
Reports
Config
API
Updater
```

Harus diuji di Termux.

80. Feature yang Dihapus

Dark Web

Remove completely.

Tidak cukup:

```
enabled = false
```

Harus dihapus dari architecture.

Alasannya:

```
maintenance cost tinggi
verification sulit
false positive risk
scope terlalu melebar
tidak menjadi core value GhostIntel
```

81. Feature yang Harus Dihapus jika Tidak Benar-Benar Terintegrasi

Prinsip:

Don't advertise what the code cannot actually do.

Jika README mengatakan:

```
GreyNoise integration
PhishTank integration
```

tetapi implementasi sebenarnya tidak ada/terputus:

hapus dari feature list, atau implementasikan dengan benar.

Jangan fake capability.

82. Module Registry

Semua module terdaftar di:

```
core/module_registry.py
```

Contoh:

```
username
email
phone
domain
subdomain
dns
ip
asn
certificate
breach
reputation
dorking
```

Registry memungkinkan:

```
enable
disable
discover
load
validate
```

secara konsisten.

83. Module Interface

Setiap module punya pola yang sama:

```
Module
├─ name
├─ version
├─ description
├─ requirements
├─ capabilities
├─ scan()
└─ health_check()
```

Dengan demikian module baru tidak merusak core.

84. Health Check

Contoh:

```
ghostintel doctor
```

Output:

```

GhostIntel Doctor

Core                OK
Database            OK
Network             OK
Telegram            NOT CONFIGURED
API Provider A      OK
API Provider B      INVALID
Dorking Engine      OK
Reports             OK
Updater             OK
  
```

```
ghostintel doctor
```

- Python
- Dependencies
- Database
- Config
- Permissions
- Network
- API
- Telegram
- Storage
- Locale
- Updater
- Modules

[██████████░░░░░░░░░░] 48%

[24/50] Checking GitHub...

Jangan:

```
Traceback...
```

langsung ke user dalam normal mode.

Normal:

```
Could not check Instagram.  
  
Reason:  
Request blocked.  
  
Use --debug for more details.
```

Debug mode baru menampilkan traceback.

88. Output Standardization

Semua module:

```
Target  
Module  
Status  
Result  
Evidence  
Source  
Confidence  
Timestamp  
Error
```

Jadi output konsisten.

89. Web UI

Web UI tidak perlu menjadi pengganti CLI.

Fungsinya:

```
Dashboard  
Scan  
Reports  
History  
API configuration  
Monitoring  
Settings  
Bug report  
Documentation
```

90. Dashboard

Contoh:

GhostIntel Dashboard

Scans

Total Scans	120
Completed	113
Failed	4
Unknown	3

Recent Activity

Username Scan

Domain Scan

Email Scan

91. Web Language

Web UI juga:

ID

EN

Switch:

[ID] [EN]

Perubahan harus global.

92. Reports Dashboard

User dapat:

View

Download

Delete

Search

Filter

berdasarkan:

date

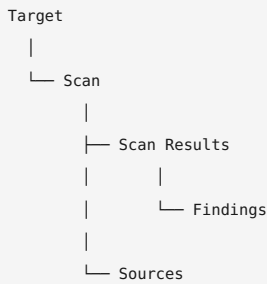
target

module

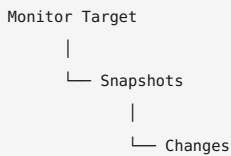
status

93. Database Relationship

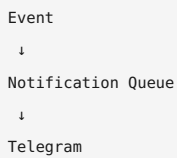
Conceptual:



Monitoring:



Notification:



94. Security & Privacy

GhostIntel harus punya:

```
Secret redaction
Safe config
No secret logging
Minimal data storage
Optional cache
User-controlled report deletion
Clear API handling
```

95. Scope Philosophy

GhostIntel harus membedakan:

Data

Apa yang ditemukan.

Evidence

Apa yang membuktikan.

Interpretation

Apa kemungkinan hubungan.

Ketiga hal ini jangan dicampur.

96. Confidence

Gunakan:

HIGH
MEDIUM
LOW
UNKNOWN

Contoh:

Username FOUND
Confidence: HIGH
Evidence: Verified profile

Sedangkan:

Possible account
Confidence: LOW

97. Jangan Overclaim

Misalnya ditemukan:

username = ruyynn

di suatu website.

GhostIntel tidak boleh otomatis berkata:

This definitely belongs to the same person.

Lebih benar:

Potential username match.
Identity correlation is unconfirmed.

98. Report Summary

Contoh:

Scan Summary

Target:

example

Platforms Checked:

50

FOUND:

12

NOT FOUND:

21

NOT ACTIVE:

4

UNKNOWN:

8

BLOCKED:

3

ERROR:

2

Duration:

00:01:42

99. File Naming

Consistent:

```
username-example-20260928-120100.json
username-example-20260928-120100.html
domain-example.com-20260928-120100.json
```

100. Logging Naming

```
ghostintel-2026-09-28.log
```

Debug:

```
debug-GI-20260928-XXXX.zip
```

101. Documentation

Docs:

```
docs/
├─ installation.md
├─ termux.md
├─ configuration.md
├─ modules.md
├─ username.md
├─ email.md
├─ phone.md
├─ domain.md
├─ subdomain.md
├─ dorking.md
├─ api.md
├─ telegram.md
├─ reports.md
├─ monitoring.md
├─ troubleshooting.md
└─ development.md
```

102. README

README harus menjelaskan:

```
What is GhostIntel?
Features
Installation
Termux
Quick Start
Modules
API
Reports
Telegram
Dorking
Configuration
Screenshots
Documentation
Contribution
License
Official Website
```

103. Official Website Integration

Di:

```
README
CLI
Help
Web UI
Reports
Documentation
```

tambahkan:

<https://www.ghintel.web.id>

Contoh footer:

GhostIntel

Official Website:

<https://www.ghintel.web.id>

104. Changelog

Gunakan:

CHANGELOG.md

Contoh:

```
## 3.5.0
```

Added

- Dorking Engine Level 1/2
- Telegram notification batching
- Doctor command
- Advanced email API integration

Improved

- Username detection
- Domain intelligence
- Termux support

Removed

- Dark Web module

Fixed

- Username false positives
- Language inconsistencies

105. Versioning

Gunakan semantic version:

3.0.0

3.1.0

3.1.1

3.5.0

4.0.0

Interpretasi:

MAJOR

MINOR

PATCH

106. Development Roadmap

Phase 1 — Foundation

Core architecture
Database
Config
Logging
Exceptions
i18n
Module registry
Request manager

Phase 2 — Reliability

Rate limiter
Cache
Proxy
API manager
Truthful result system
Evidence
Confidence

Phase 3 — Core Modules

Username
Email
Phone
Domain
DNS
IP
ASN
Certificate
Subdomain

Phase 4 — Dorking

Dorking Engine
Level 1
Level 2
Template system
Export

Phase 5 — Reporting

JSON
Markdown
HTML
TXT

Phase 6 — Telegram

- Bot
- Events
- Queue
- Rate limit
- Batching
- Deduplication

Phase 7 — Debug & Bug Reporting

- Debug mode
- Crash handler
- Bug ID
- Debug bundle
- Bug report system

Phase 8 — Monitoring

- Target monitoring
- Snapshots
- Diff
- Notifications

Phase 9 — Web UI

- Dashboard
- Scan
- History
- Reports
- API
- Settings
- Bug report

Phase 10 — Updater

- GitHub version check
- Manifest
- Checksum
- Backup
- Update
- Rollback strategy

Phase 11 — Testing

- Unit
- Integration
- E2E
- Termux
- CI

107. Prioritas Development

Kalau kita benar-benar mulai coding, urutannya menurut gw:

1. Core architecture
2. Database
3. Config
4. Request Manager
5. API Manager
6. Rate Limiter
7. i18n
8. Findings schema
9. Username engine
10. Domain engine
11. Email engine
12. Phone engine
13. Subdomain/DNS/IP
14. Dorking Engine
15. Report Engine
16. Telegram
17. Debug/Bug Reporter
18. Monitoring
19. Web UI
20. Updater
21. Full testing
22. Documentation

Jangan langsung mengejar Web UI.

Fondasi harus stabil dulu.

108. Definition of Done

Sebuah module tidak dianggap selesai hanya karena:

command bisa dijalankan

Module baru dianggap selesai jika:

- ✓ Function works
- ✓ Error handling works
- ✓ Rate limit works
- ✓ Timeout works
- ✓ Logging works
- ✓ i18n works
- ✓ Database integration works if needed
- ✓ Report integration works
- ✓ Test exists
- ✓ Termux compatibility checked
- ✓ Doesn't leak secrets

109. Definition of Done untuk API

API integration dianggap selesai kalau:

- ✓ API key can be configured
- ✓ API key is securely stored
- ✓ API connection can be tested
- ✓ API is actually called
- ✓ Response is parsed
- ✓ Errors are handled
- ✓ Rate limit is handled
- ✓ Result enters standard Findings schema
- ✓ Report contains source
- ✓ Tests exist

110. Definition of Done untuk Username

Username engine dianggap selesai kalau:

- ✓ Platform adapters
- ✓ FOUND detection
- ✓ NOT_FOUND detection
- ✓ UNKNOWN handling
- ✓ BLOCKED handling
- ✓ TIMEOUT handling
- ✓ Rate limiting
- ✓ Deep profile metadata
- ✓ Evidence
- ✓ Confidence
- ✓ Preview limit
- ✓ Full report
- ✓ Tests

111. Definition of Done untuk Update

Updater dianggap selesai kalau:

- ✓ Checks official source
- ✓ Compares versions
- ✓ Detects update
- ✓ Shows changelog
- ✓ Verifies download
- ✓ Checks checksum
- ✓ Protects user config
- ✓ Protects database
- ✓ Handles failure
- ✓ Supports manual update
- ✓ Has no-update message

112. Final GhostIntel Feature Map

Core

- ✓ Engine
- ✓ Config
- ✓ Database
- ✓ Request Manager
- ✓ API Manager
- ✓ Rate Limiter
- ✓ Cache
- ✓ Proxy
- ✓ Logging
- ✓ Debug
- ✓ Bug Reporter
- ✓ i18n
- ✓ Module Registry
- ✓ Findings
- ✓ Correlation
- ✓ Monitoring
- ✓ Notification
- ✓ Updater

OSINT Modules

- ✓ Username
- ✓ Email
- ✓ Phone
- ✓ Domain
- ✓ Subdomain
- ✓ DNS
- ✓ IP
- ✓ ASN
- ✓ Certificate
- ✓ Breach
- ✓ Reputation
- ✓ Dorking

Output

- ✓ CLI
- ✓ HTML
- ✓ JSON
- ✓ Markdown
- ✓ TXT

Integrations

- ✓ API Providers
- ✓ Telegram
- ✓ Official GitHub
- ✓ Official Website

Platforms

- ✓ Windows
- ✓ Linux
- ✓ Termux

Quality

- ✓ Unit Tests
- ✓ Integration Tests
- ✓ E2E Tests
- ✓ Fixtures
- ✓ Error Handling
- ✓ Debugging
- ✓ Health Check

113. Hal yang Sengaja Tidak Kita Kejar

Supaya GhostIntel tidak berubah menjadi project yang punya 100 fitur tapi 60% tidak stabil, untuk sekarang **jangan menambahkan fitur random** seperti:

```
AI chatbot
face recognition
mass scraping
random vulnerability scanner
exploit automation
credential harvesting
dark web crawler
```

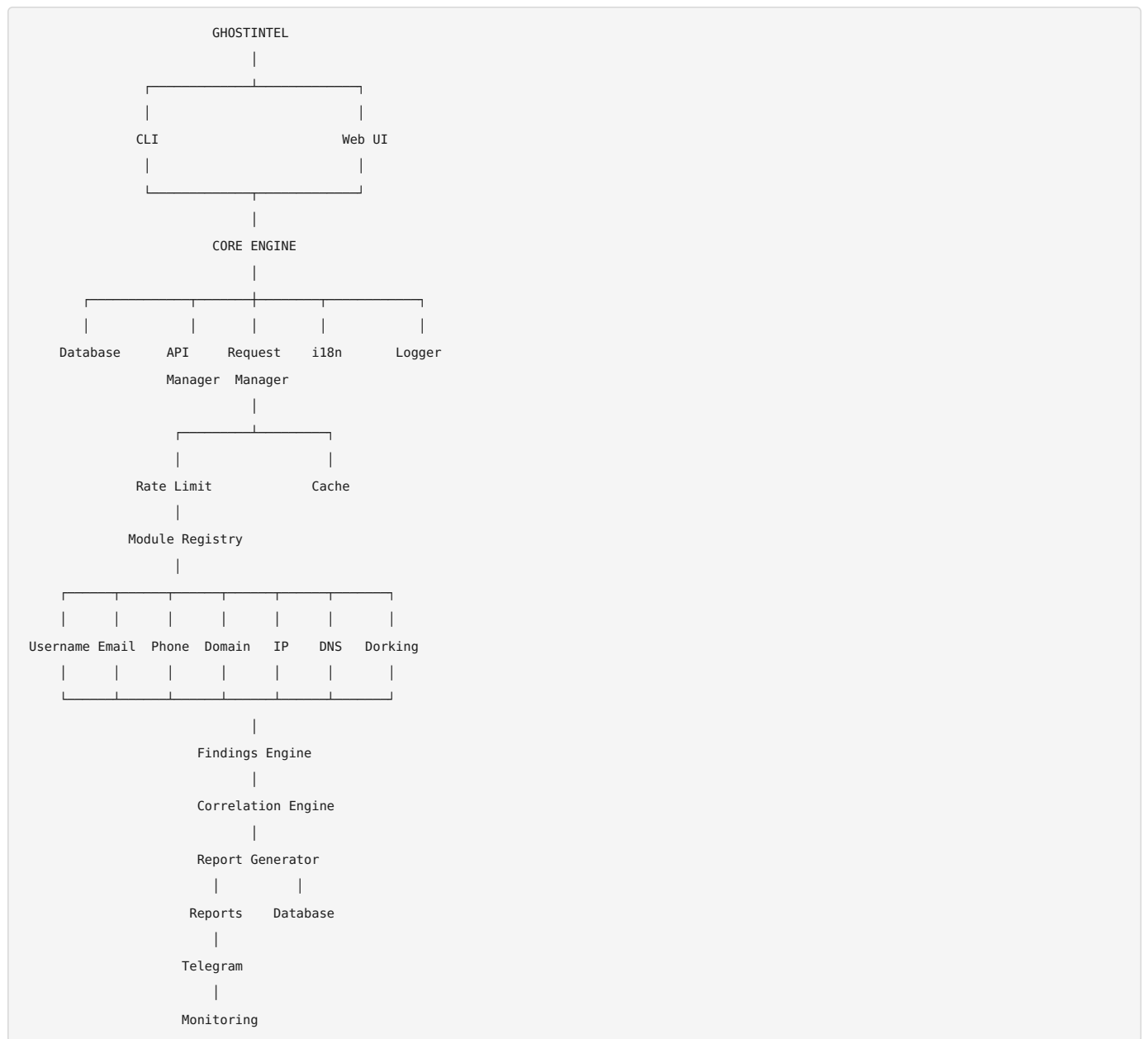
Kalau suatu saat memang diperlukan, kita evaluasi sebagai project/module terpisah.

GhostIntel tetap fokus:

```
OSINT collection + validation + correlation + reporting.
```

114. Blueprint Singkat Arsitektur Final

Pada akhirnya GhostIntel akan punya alur seperti ini:



115. Target Akhir

Kalau blueprint ini diimplementasikan dengan benar, GhostIntel 3.x bukan cuma:

“tool untuk mencari username.”

Tetapi menjadi framework dengan lifecycle:

```
INPUT
↓
DISCOVERY
↓
VALIDATION
↓
ENRICHMENT
↓
CORRELATION
↓
EVIDENCE
↓
REPORT
↓
MONITOR
↓
NOTIFY
```

Dan yang paling penting:

```
Discovery ≠ Proof
Possible ≠ Confirmed
Unknown ≠ Not Found
HTTP 200 ≠ Account Found
API Configured ≠ API Used
Feature Listed ≠ Feature Implemented
```

Itu yang menurut gw harus menjadi **DNA GhostIntel 3.x**.

Satu hal lagi yang menurut gw penting: karena repository GhostIntel publik saat ini memang masih tercatat sebagai framework Python OSINT dan diperbarui pada 3 Agustus 2026, blueprint ini paling masuk akal dijadikan **target architecture untuk refactor/major iteration**, bukan sekadar menumpuk patch di atas struktur lama. :chatgpt-content-reference{index="3"}

Dan untuk bagian Help yang lo minta, kita kunci sebagai **official website footer kecil di bawah banner**:

```
GhostIntel v3.x
OSINT Intelligence Framework

Official Website:
https://www.ghintel.web.id

Visit the official GhostIntel website for documentation,
updates, guides, and the latest information.
```

Untuk versi Indonesia:

```
GhostIntel v3.x
OSINT Intelligence Framework

Official Website:
https://www.ghintel.web.id

Kunjungi website resmi GhostIntel untuk dokumentasi,
update, panduan, dan informasi terbaru.
```

Ini blueprint master-nya.